

CHAPTER 1

INTRODUCTION

TriBrid vehicles represent a groundbreaking shift in automotive technology, combining the strengths of electric motors and traditional internal combustion engines to provide a more efficient and eco-friendlier alternative to conventional vehicles. By utilizing both power sources, Tri Brid vehicles are able to optimize fuel consumption, reduce harmful emissions, and provide greater flexibility and performance in diverse driving conditions. These vehicles represent a crucial step forward in the ongoing effort to reduce the environmental impact of transportation, offering a cleaner and more sustainable mobility solution. The importance of sustainable energy solutions in the modern world is critical as the negative impacts of fossil fuel consumption on the environment become increasingly evident. With transportation contributing significantly to global carbon emissions, adopting greener alternatives is essential in mitigating climate change. The shift from fossil fuels to renewable energy sources, such as solar and electric power, has become a focal point for innovation in the transportation sector. By transitioning to cleaner energy, we can reduce the reliance on non-renewable resources, decrease pollution, and pave the way for a more sustainable future. The integration of solar power into vehicles is one of the most promising solutions for addressing energy consumption and environmental concerns. Solar panels can capture energy from the sun and convert it into electricity to power the vehicle's electric motor or recharge the battery. This system allows the vehicle to run more efficiently by utilizing a free and renewable energy source, especially in sunny regions, reducing the need for frequent recharging from the grid.

Furthermore, integrating solar power into Tri Brid vehicles enhances their energy autonomy, allowing them to operate with minimal reliance on external charging infrastructure.

This project aims to design and develop a Solar Power Car with Tri Brid Petrol, LPG, and Electric Systems, enhanced by a self-recharge system through an alternator. The vehicle will intelligently switch between power modes electric, LPG, and petrol based on speed and driving conditions, ensuring optimal fuel efficiency. The self-recharge feature will enable the battery to be replenished during vehicle operation, reducing the need for external charging stations. Additionally, a ESP8266-based control system will manage energy switching, monitor system performance, and enable live tracking via IoT technology. This project seeks to offer an advanced, sustainable, and adaptable solution to future transportation challenges while demonstrating the potential of solar energy in Tri Brid vehicle applications. Through this innovative approach, the project not only contributes to environmental sustainability but also sets a foundation for future developments in smart, solar-powered Tri Brid vehicles. The use of alternative energy sources, self-recharge capabilities, and intelligent control systems demonstrates the potential of combining cutting-edge technology with sustainable practices to revolutionize the way we think about transportation.

LITERATURE SURVEY

Title 1 : Solar-powered electric car: validity and effectivity of prop inenergy conversion learning

Journal Title : Solar-powered electric car: validity and effectivity of prop in energy conversion learning

Authors : Rahmat Rizal, Haji Aripin, I Made Joni

Year : 2024

Learning resources in the learing environment. The maximum achievement of learning objectives can indicate practical learning activities and cannot be separated from how the learning process is carried out inside and outside the classroom. The goal of physics learning activities is to facilitate students' understanding of the concepts and principles of physics. It is expected to be the basis for developing knowledge,skills, and self-confidence. It can utilize all aspects obtained during learning to be implemented in daily life. Physics learning focuses on learning objectives that lead to the embodiment of the physics concept. It is oriented toward finding truth through a systematic and holistic thought process without ignoring norms related to humanity. With technological advances that have developed significantly, physics learning activities benefit from facilitating learning activities through technology-based learning media, innovation, and creativity. Effective physics learning cannot be separated from the media. Effective learning media really has a significant impact on learning activities. Media, which includes learning aids in teaching, also means the carrier of messages or learning resource information to recipients of learning messages. As transmitters and distributors of messages, learning media can represent teachers in presenting learning material to students. The use of learning media in the teaching and learning process can generate new desires and interests, create

motivation, stimulate teaching activities, and even have a psychological effect on students. Rizal states that the benefits of learning media include i) making the learning process more interesting so that it can raise students' learning motivation, ii) clarifying learning material so that students can easily understand the material, iii) making the learning process more varied, and iv) involving students in carrying out a number of activities during learning, including observing, performing, demonstrating. Learning media that is used effectively in various forms can help reduce visual perception difficulties experienced by students. Energy conservation is one of the courses in the physics education department at Universitas Sili Wangi. This course supports the achievement of competency in attitudes and analysis in the field of energy utilization and management, especially in the conversion and conservation sectors. With a good understanding of the use and management of energy, it is hoped that we can meet human needs in the present without reducing the ability to fulfil energy in the future (energy sustainability) and still pay attention to ecological aspects to achieve balance. To be able to hold this lecture, students require independent experimental activities or interactive demonstrations to present information related to energy conversion. The physics education laboratory facility currently consists of 1 basic physics laboratory room with a practicum capacity of 20 people. It equips with practicum tools in the form of 4 units of basic physics practicum of Integrated Instrument Components (IIC) for each central practicum theme (mechanics, electricity and magnetism, optics, and waves and thermodynamics). The current practicum IICs cannot show contextual energy conversion with renewable energy sources. Implementing laboratory activities still faces various obstacles, such as incomplete tool sets, unavailable tools and materials, inadequate activity instructions, damaged laboratory equipment. The limitations of the available tools to support

physics learning will have an unfavorable impact on learning outcomes. Considering the need for several instruments to show the phenomenon of energy conversion that is more contextual, it is necessary to have an innovative learning medium in the form of an energy conversion demonstration instrument. The props are one of the learning media, which is a form of depicting the working mechanism of an object. A prop is a tool used to educate and deliver material, either objects or behaviour, making it easier for students to understand the material. The props contain the characteristics and forms of the teaching materials used to demonstrate the material in events and activities so that the material can be more easily understood. Using props in the learning process is one way to support the development of knowledge, skills, and basic needs for delivering material, concepts, and physics information. It can be used as learning media and channel messages that stimulate students' thoughts, feelings, and willingness to encourage their learning process. Educational visual props are arranged based on the principle that the knowledge that exists in every human being is received or captured through the five senses. The props have a function to demonstrate events, activities, phenomena, or the working mechanism of an object. The existence of props is intended to mobilize as many senses as possible to an object to facilitate perception. Seeing the potential of teaching instruments that can support the effectiveness of learning activities and renewable energy issues, the research developed a solar electric car, which will be used as a teaching instrument in energy conversion learning. One of the issues often present in energy conservation is the need for renewable energy to overcome the increasing energy needs and replace the diminishing fossil energy sources.

In a press release, the ministry of energy and mineral resources stated that in a global culture, the world would continue to move towards utilizing energy sources by reducing fossil energy and switching to environmentally friendly energy. The demand for green products produced by the green industry is increasing and even becomes a necessity if you do not want your products to be subject to a carbon border tax at the global level. Green products were also reinforced by the President of the Republic of Indonesia in his mandate, prioritizing himself in initiating energy transformation towards new and renewable energy. A green economy, green technology, and green products must be strengthened so that we can compete in the global market. Learning props on energy conservation and renewable energy requires the development of teaching props that adapt to these two needs. Several researchers have developed teaching props used in learning energy conversion. Pauji has developed an energy conversion tool for high school students. The toolset converts motion energy into electrical energy using a dynamo. Jufri developed a simple energy conversion tool that can only show the change in the form of motion energy into electrical energy, which is focused on elementary science learning. Rohman has developed a demonstration tool for converting energy from various forms of energy into electrical energy. These props are large and complex to mobilize. From all of this research, the props that have been developed do not show a display of magnitude energy before and after conversion, so students do not find out how efficient the props are. Apart from that, students only use tools indoors. Considering the shortcomings of existing energy conversion props, it is necessary to develop a learning prop that can provide memorable learning experiences to students and also consider the use of renewable energy.

Title 2 : Study of a New Type of Electric Car: Solar-Powered Car
Journal Title : Study of a New Type of Electric Car: Solar-Powered Car
Authors : Muhammad Umer, Lars Kulik, Egemen Tanin.
Year : 2021

Solar energy is becoming more and more important in recent years, as an alternative to fossil fuels. Solar energy is also one of the renewable clean energy resources. It can be used to generate electric energy into the car via solar panels. Also, there are no pollution and greenhouse gases released when solar energy is converted into electric energy. This new transportation technology will improve environmental problems effectively. This world's first long-range solar-powered electric car called Lightyear One was produced on 25th June 2019, which was predicted to start the production in 2021. This essay analyses the trend of whether solar-powered cars are possible to be widespread and to take first place among different types of electric cars in the future. One obvious difference between solar cars and electric cars is the initial energy resource. Solar cars use electricity that is transformed directly from sunlight, which can be achieved by using solar panels consisting of many individual solar cells. In the following picture, which is the image of Lightyear One, we can see that the solar panels have covered its whole roof and hood consisting of 16 square feet in total, to charge the car, whenever there is light radiation shining on the top. The solar panels, consisting of many solar cells, can absorb the light radiation to generate the electric energy which can be stored in the car's accumulator so that a large amount of electricity is saved for recharging the cars. The importance of solar cells, moreover, can be found especially when the electric energy completely runs out, in that scenario, the cell-powered energy storage can continue supporting the car

to drive for a long range without recharging. Therefore, the solar panels on the roof and hood of the cars are used for not only an increasing driving distance but also to decrease the amount of fossil fuels burnt for providing the electricity. Without a doubt, it is possible for solar-powered cars to have a total, final amount of energy is much larger than the initial energy despite already driving for a while. Compared with the pure electric cars, the solar-powered cars may reach a level of zero carbon dioxide emission in a strict sense. While the pure electric cars do not. In fact, all-electric cars do not have tailpipes, so obviously, there is no pollution released as they are driven. However, the main which causes only a small portion of people who desire electric cars is the limitation of driving distance due to a small electric energy storage capacity. This means that consumers have to look for a charging station to power their cars with enough electricity after driving for a period. Most of the electric energy supplied by the charging stations is provided through burning the fossil fuels. As a result, with the increase of electric energy used and a growing number of populations driving electric cars, more fossil fuels will be burned to supply the electric energy. Which will cause more carbon dioxide to be released from the heat power plants across the world. However, if it is possible to achieve the production of solar-powered cars, the problem that there is an increasing amount of carbon dioxide emission will not happen anymore. This also means the environmental problems caused by global warming will be improved. It is clearly seen that the carbon dioxide emission in China is the most, and the environmental problem is still serious in America. Considering the positive influence on the environment, it is speculated that the China and America might think about carrying out solar-powered cars firstly. Solar energy is the fastest growing energy resource and it has supplied about 2% of the electric energy in the whole world. Even though that sounds not too much, it represents a 50-fold increase over the past decade. We can calculate the

sun's insolation to show the solar energy is sufficient for human's daily use according to the following equation. When it comes to solar cars, solar panel efficiency is the most worrying problem among the public. According to a journal reported in Boston Solar Blog, in fact, the efficiency of solar panels depends on the temperature and the shaded area. Generally, there is a peak solar efficiency between 59 Fahrenheits and 95 Fahrenheits (which is between 15 Celsius and 35 Celsius.) If the temperature of the solar panels continues rising up to 144 Fahrenheits which is about 62 Celsius, the efficiency will be declined a little bit. However, all countries in the world are hardly ever hotter, another problem is that how much the shade or cloudy weather can affect the amount of electric energy stored. As a general rule, the energy produced through the solar panels under the clouds and the shades is as half as that under the sunny days. The shade caused by the trees or high buildings on the roof or hood of the car, where the solar panels are installed, will disappear after a period because the earth is rotating around the sun all the time, so there would not always be shade on the solar panels. Besides, the cloud changes its place any time no matter when the car is stopping on the road or a person is driving it, there are always some gaps that allow the sunlight to irradiate on the top of the solar panels. This amount of energy is sufficient for a short journey. According to the announcement from the Lightyear company.

we can know the driving mileages of Light-year One is about 450 miles (725km) with one single charged battery, which is much longer than the other electric cars (compared to the tesla model S's 379 miles). And on average, the mileages can increase by about 7.5 miles per hour only converting the solar energy when the owners just stop their cars outside. Therefore, by assuming the solar panels of the cars working for eight hours

per day, the total mileages will increase by 60 miles per day. which is enough for those people who drive their cars for a short journey. When it comes to the total distance traveling only based on solar energy, it is assumed 60 miles increased per day will only be powered for 100 days in a year, only 100 days can get to 60 miles per day by solar energy, which means there will be about 6000-mile range only powered by solar cells in a year. As a result, regardless of the charged electricity, this range in miles can be satisfied with the request for short traveling, what's more, there will be no extra money paid for the electricity. Firstly, because solar energy is a renewable and clean energy resource, solar energy will be paid more attention by the government in the future rather than the traditional fossil fuels. This means installing the solar panels on the roof and hood of the cars is an expected trend. Secondly, comparing with other types of electric cars, such as pure electric cars, the range of the solar-powered car is much larger, and it also can continue being increased by the solar cells.

Title 1 : A Study and Design Based Simulation of Hybrid Solar Car

Journal Title: A Study and Design Based Simulation of Hybrid Solar Car

Author : Ranjeet Singh , M.K.Gaur,Chandra Shekhar Malvi

Year : 2013

The brief study of Tri Brid solar car is efficient in our daily life because now day's pollution and fuel rate is very big problem many people having fuel cars. Use of solar energy is being used for car, besides the control of vehicular pollution in the city, less consumption of fuel, Tri Brid solar car are effective reducing global warming and environment problem in big frame. In the present work, the objective of this work is to estimate the potential of both energy as PV energy and mechanical engine power, both powers will be utilized in running car with weight reduction can be achieved primarily by the introduction of better material, design optimization and better manufacturing processes. The Tri Brid solar car is one of the potential items for weight reduction in Tri Brid solar car as it accounts for 5% - 10% of the weight. Various advantages for Tri Brid solar car by using solar technology.

1. Reduction in conventional car demand in urban city
2. Minimum the pollution problem in urban city
3. Give clean energy which will reduce the carbondioxide emission every month.
4. Reduction in fuel demand.

A Tri Brid car is a vehicle which can be used three power sources are a solar energy with electric motor, electrically operated and a small combustion engine to run a car. They are slowly gaining popularity with the auto buyers because they are seeing the benefits that owning a car will help them to reduce their carbon emission and is also energy efficient. It is also healthful for

environment. In this paper, we are firstly creating a drawing and design of Tri Brid solar car by using CATIA V5R19 tool after that we performed using the finite element method (FEM) done using ANSYS 13.0 software. Modelling was done for front and rear unit frame 3D brick element (solid 45) and five-node. Battery bank is an important component for the HSC. It has been generated 24 V DC for supplying to the electric motor and also electronic devices in the HSC. IC engine has been employed for driving of Tri Brid solar car which can be used during a night or minimum electric energy of HSC. Electronic controlling unit (ECU) is an electronics circuit that is used for controlling the energy in the electric motor which can be provides a speed variation.

1. Front drive system – The IC engine is placed on the front of the car, With the help of chain drive mechanism the power stored in crank shaft is driven to the axle of front wheels.
2. 2. Rear drive system – The electric motor is placed near the rear axle of Tri Brid solar car which can be operated by charged batteries then electric motor rotate the rear axle of rear wheels Both systems are placed in Tri Brid solar car, at a onetime one system can be used in Tri Brid solar car, at a onetime one system can be used in transmission of power to wheels as either rear or front wheels of a Tri Brid solar car If one system gets fail then we have an option to choose from alternative driving mechanism.

Title 4 : Solar Energy based Tri Brid Electric Car
Journal Title: A Study and Design Based Simulation of Hybrid
Authors : Dr. Siva Ganesh Malla, Priyanka Malla, Rjesh Koilada
Year : 2010

Global warming is a very serious concern in worldwide with the increasing temperature every year and increasing number of gases causing global warming like carbon dioxide, Sulphur, etc. One of the main sources for these gases are the emissions from vehicles running on conventional fuel such as petrol, diesel etc. As the global awareness on pollution and face of climate change, the researchers are researching into alternative sources of energy [1-2]. Since automobiles are currently a major source of air pollution, governments and major automotive companies are collaborating to provide an effective solution that will result in the reduction of carbon emissions by reducing the consumption of fossil fuel. The best solution to reduce the emission of carbon from vehicles is to develop the electric vehicles [2]. Since the late 19th century until recently several electric vehicles have been designed, manufactured and used throughout the world. Some developments are just prototypes, some are just for concept models of cars, and others were just special purpose vehicles. Electric cars with energy sources as lead-acid batteries are developed for locomotion since the end of the 19th century.

These batteries were replaced by the NiMH and Li-ion types later, but in terms of the price with energy ratio, till lead acid batteries still rule. Therefore, it is common to find lead-acid batteries are used in electric cars. In the present world, some of Tri Brid electric cars are having Diesel engines. These cars have a price and environmental disadvantage, as Diesel engines are more expensive as well as causes to increase

pollution. Apart from the Tri Brid vehicles using engines, the Tri Brid vehicles also deal only with electric motors. In these types of electric vehicles are having only motor as mechanical energy source. Both the alternating current (AC) and direct current (DC) motors are suitable to use in electric vehicles with the help of proper controllers. The selection of motor will be depended on many factors like size, cost, performance, power losses in motor, controllers, etc. Both types of motors are used in electric vehicles and have their own advantages and disadvantages. This electric motor and their models in presently existing electric vehicles (EEV) are induction motors, brushless DC motor, permanent magnet synchronous motor (PMSM) and switched reluctance motor (SRM). In electric vehicle, the control objective is the torque of the driving machine. The control system is required to be fast responsive and low-ripple in torque. EV requires that the driving electric motor has a wide range of speed regulation. By considering the AC motors we are reducing the size of the machine and cost of the equipment. The main drawback is the limitation of energy provided by the battery in electric vehicles. The distance travelled by car will be depends on availability of power in the battery. Hence, an alternate charging patch for batteries is investigated in this paper. The photovoltaic cells are most effective solution to charge the batteries even in running condition of the car. The combination of battery and PV system will help the car to cover the more distance. This type of electric vehicles can be called as Hygienically Electric Vehicles (EVs) or Tri Brid Electric Vehicles (TEVs) are having body of the vehicle, Motor, Power supply from batteries, Power converters and controlling equipment. Apart from these, we have to assume the average weight of the passengers which can capable to drive by the vehicle. In this paper, consider the solar panels also for supply the power to batteries as well as motor. Hence, we included the weight of the above-mentioned equipment while calculating the rating and

sizing the devices or components. The body weight of the vehicle we considered from reference. According to the references and by considering the weight of engine, assumed the weight of the body is approximately 1500 Kg. Hence, the calculation of the size and ratings of the components. For deciding the required power rating of the motor for proposed Tri Brid electric vehicle, the vehicle dynamics like rolling resistance, gradient resistance, aerodynamic drag, etc. has to be considered. For illustration procedure 1500 Kg of gross weight is considered for Tri Brid electric vehicle.

CHAPTER-3

PROJECT METHODOLOGY

3.1EXISTING SYSTEM

The existing systems for Tri Brid vehicles primarily focus on combining traditional internal combustion engines (ICE) with electric motors to improve fuel efficiency and reduce emissions. While some advanced electric vehicles (EVs) use a pure electric motor, Tri Brid systems incorporate both gasoline, diesel, or LPG engines to provide an efficient balance between power and fuel consumption. The key components of an existing Tri Brid system include.

3.1.1 Tri Brid Powertrain System

Tri Brid vehicles typically use a combination of an electric motor and a traditional internal combustion engine (ICE). The powertrain of a Tri Brid vehicle is designed to intelligently switch between the electric motor and the engine based on driving conditions, optimizing fuel consumption and performance. The system is controlled by a **Battery Management System (BMS)**, which ensures efficient use of power from both sources. The main types of Tri Brid systems are,

1. **Series Tri Brid:** In this system, the engine generates electricity to charge the battery, which then powers the electric motor.
2. **Parallel Tri Brid:** Both the engine and the electric motor can drive the wheels simultaneously or independently, depending on driving conditions.
3. **Series-Parallel Tri Brid:** This system combines the features of both the series and parallel Tri Brid s, allowing for greater flexibility in power distribution.

Example: Toyota Prius and Honda Insight are well-known examples of Tri Brid vehicles using such systems.

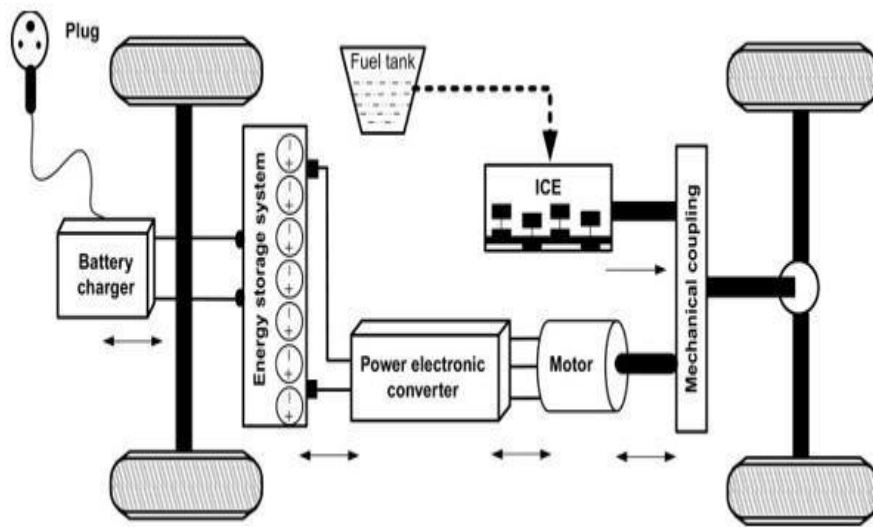


Figure 3.1 Tri Brid Powertrain System

3.1.2 Solar-Powered Vehicles

Solar-powered vehicles are designed to reduce the dependency on external power sources, using solar panels to generate electricity from sunlight. The vehicle's solar panels provide energy for auxiliary power systems, such as the air conditioning and battery charging, or directly power the electric motor for low-speed driving. Solar vehicles are typically limited in range and performance due to the relatively low power output of solar panels compared to the energy requirements of a moving vehicle.

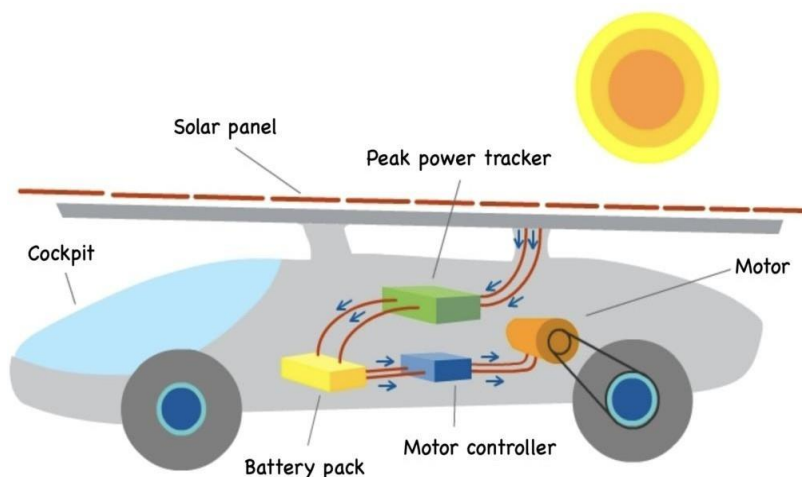


Figure 3.2 Solar-Powered Vehicle

3.1.3 Self-Recharge and Regenerative Braking Systems

Many Tri Brid and electric vehicles employ regenerative braking systems, which convert the kinetic energy generated during braking into electrical energy. This energy is used to recharge the vehicle's battery, enhancing the overall efficiency of the vehicle. Regenerative braking can be particularly useful for vehicles in city driving conditions, where frequent stopping and starting occur. In addition to regenerative braking, some Tri Brid vehicles use **alternators** to generate electricity while the engine is running, helping to recharge the battery without external charging sources. This allows the vehicle to maintain battery charge during normal operation, reducing the need for frequent recharging.

Example: Tesla Model S and Nissan Leaf are electric vehicles that integrate regenerative braking systems for energy efficiency.

3.1.4 Intelligent Control Systems

Existing Tri Brid vehicles often feature **intelligent control systems** that manage the interaction between the electric motor and internal combustion engine. These systems determine when to switch between energy sources based on driving conditions, such as speed, acceleration, and load. The control systems may include **ECUs (Electronic Control Units)**, **Battery Management Systems (BMS)**, and **Energy Management Systems (EMS)** that optimize fuel consumption and battery usage. Many electric vehicles also use **smart charging systems** that enable the battery to charge from various external sources (grid, solar, or regenerative braking). These systems are often integrated with **IoT-based applications** for real-time monitoring of energy consumption, battery health, and driving performance.

Example: The **BMW i3** and **Chevrolet Volt** use smart control systems to optimize energy usage between their electric motor and petrol engines.

Challenges in the Existing Systems:

Limited Solar Efficiency:

The main limitation of current solar-powered vehicles is the relatively low efficiency of solar panels, which can only generate a small fraction of the total energy required to power the vehicle. This makes solar power an auxiliary source, with grid charging or alternative fuel sources still needed for long-distance travel.

Energy Management:

Tri-Brid vehicles with multiple power sources need sophisticated energy management systems to balance the distribution of power and ensure optimal fuel efficiency. Current systems require complex algorithms and hardware, and the challenge remains in creating more efficient and seamless switching between power sources.

Range Limitations:

While Tri-Brid vehicles extend the range of electric cars by using an internal combustion engine, the range of solar-powered vehicles is still relatively limited due to the power constraints of solar panels. They typically function well only in sunny conditions and for short trips.

Cost and Complexity:

The integration of Tri Brid systems, solar panels, regenerative braking, and intelligent control systems increases the cost and complexity of the vehicle, making it less accessible to a wide consumer base.

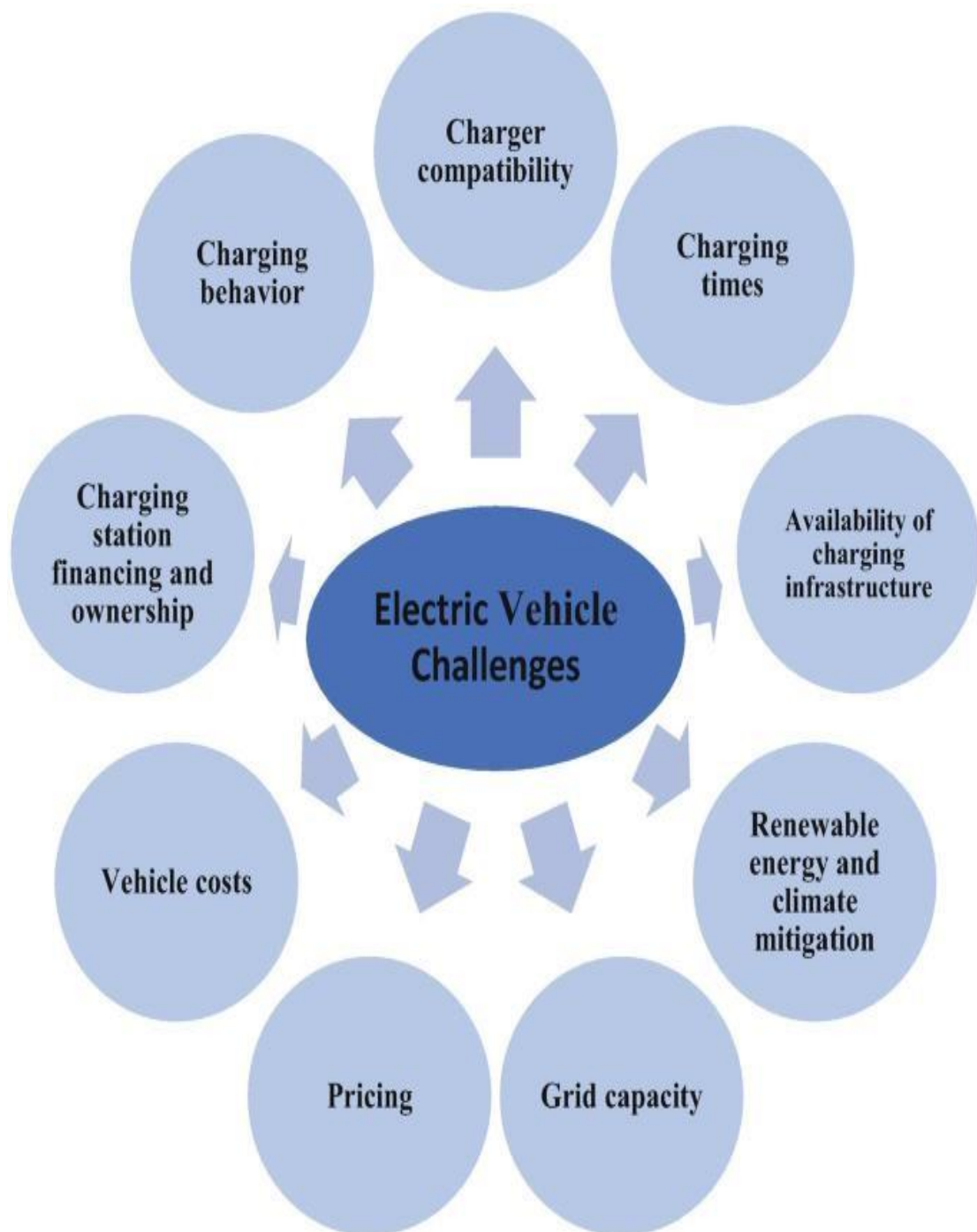


Figure 3.3 EV Challenges

3.1 PROPOSED SYSTEM

Solar Power Car with Tribrid Petrol, LPG, Electric, by Self-Recharge System
The proposed system aims to integrate solar power, Tribrid petrol, LPG, and electric power, and self-recharge mechanisms to develop a highly efficient, environmentally friendly, and autonomous vehicle. The system will combine advanced energy management techniques with sustainable energy sources to optimize fuel consumption, reduce emissions, and extend vehicle range, while offering a smart, self-recharging solution.

3.2.1 Tribrid Powertrain System

The proposed system will feature a Tri Brid powertrain, combining three power sources

Electric Motor:

Used primarily for low-speed driving or short distances, powered by the vehicle's battery. The electric motor will ensure the vehicle operates without emissions when driving at low speeds (less than 20 km/h).

LPG Engine:

Activated when the vehicle is traveling at medium speeds (between 20-50km/h). LPG is a cleaner alternative to traditional petrol and will help reduce emissions compared to gasoline-powered engines.

Petrol Engine:

Engaged when the vehicle reaches high speeds (above 50 km/h). The petrol engine will provide maximum power for long-distance travel, ensuring the vehicle can operate on highways without limitations. The vehicle will have an intelligent Energy Management System (EMS) that will decide the optimal power source to use based on speed, load, and driving conditions. The system will switch seamlessly between power sources to ensure fuel efficiency and optimal performance.

3.2.2 Solar Power Integration

The solar power system will be integrated into the vehicle's design, with photovoltaic (PV) panels installed on the roof or body. The solar panels will generate electricity during daylight hours and recharge the battery for the electric motor or supply power to auxiliary systems such as air conditioning, lights, or the battery charging circuit.

The primary goal of the solar panels is to:

Extend the range of the electric vehicle by providing supplementary charging. Reduce reliance on grid electricity for recharging the battery, making the vehicle more independent and eco-friendlier. Provide a renewable energy source, ensuring that the vehicle remains sustainable by utilizing free, abundant sunlight.

3.2.3 Self-Recharge System (Reverse Charging via engine coil)

The proposed system will incorporate a self-recharge mechanism through the use of a dynamo . This system will Generate electricity while the vehicle is in motion, when the engine is running.

Recharge the battery when the vehicle is in use, reducing the need for external charging stations and allowing for better energy autonomy. The alternator will work in tandem with the petrol and LPG engines, providing additional power to charge the battery or supply the electric motor when needed. This reverse charging mechanism ensures that the vehicle can be continuously recharged without the need for frequent stops at charging stations, making it suitable for long-distance travel.

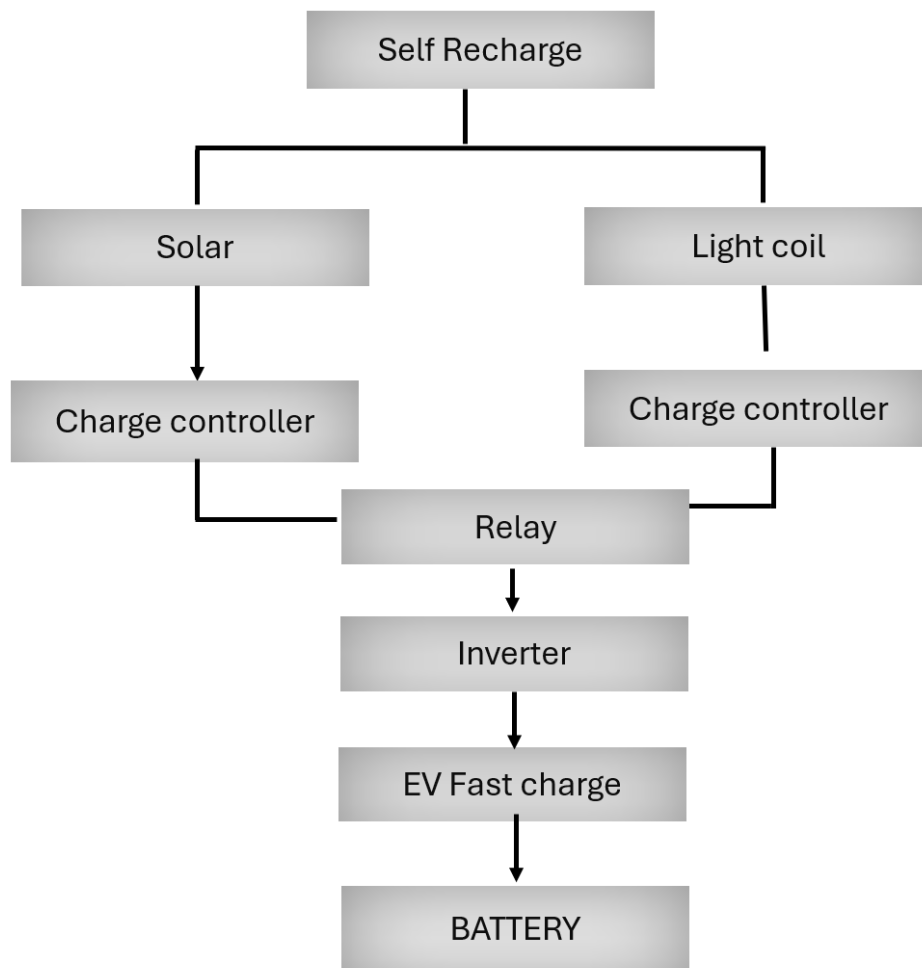


Figure 3.4 Self-Recharge System Flowchart

Flow chart Explanation:

The flowchart illustrates a self-recharging system designed to power a battery using two renewable energy sources: solar energy and a light coil. At the top level, the system initiates with the concept of "Self Recharge," which draws energy from both a solar panel and a light coil. The solar panel captures sunlight and converts it into electricity, while the light coil—likely a dynamo or electromagnetic generator—produces electricity from motion or vibrations. Each of these sources is connected to its own charge controller. These charge controllers play a critical role in regulating the voltage and current from their respective energy sources to ensure safe and efficient charging.

The outputs of both charge controllers are then fed into a relay. The relay acts as a control switch, managing the combined flow of electricity from the solar and light coil systems. From the relay, the electricity flows into an inverter, which converts the DC (Direct Current) output from the sources into a form suitable for fast charging—typically AC (Alternating Current) or high-voltage DC, depending on the system requirements. This converted power is then directed into an EV fast charge module that ensures rapid and optimized charging of the battery. Finally, the battery stores this energy for use in powering the electric vehicle or any other connected load. This system enables continuous self-charging by utilizing multiple energy sources to maximize efficiency and extend battery life.

3.2.4 Intelligent Control System (ESP8266-based)

frequent stops at charging stations, making it suitable for long-distance

The ESP8266-based control system will act as the central brain of the proposed system, managing and optimizing the energy distribution between the power sources. The system will Monitor and control power switching between electric, LPG, and petrol modes based on real-time driving data, such as speed, acceleration, and battery charge level. Enable reverse charging from the alternator and solar panels, optimizing energy storage and consumption. Monitor vehicle health (battery status, motor performance, fuel efficiency, etc.) and provide real-time diagnostics. Use IoT technology to integrate a live GPS tracking system and enable live vehicle monitoring via a mobile application, allowing users to monitor vehicle performance and energy consumption remotely. Display essential data on LCD screens or through a mobile app, allowing users to easily track their vehicle's performance, battery charge, energy consumption, and fuel levels. Automatic alerts for vehicle status changes, battery health, and fuel switching will notify the driver to ensure the smooth operation of the system.

3.2.5 IoT-Based Live Monitoring and Tracking

Incorporating IoT technology into the system will allow for live vehicle tracking and monitoring of various parameters. GPS-Based Tracking: The system will include a GPS module to track the location of the vehicle, providing real-time data on the vehicle's position and route. Live Vehicle Monitoring: Using a mobile app or web interface, users can view real-time data on energy usage, battery health, power source status (electric, LPG, petrol), and even track fuel efficiency and system diagnostics. Cloud Integration: Data collected from the vehicle will be stored in the cloud for further analysis and optimization. It will also provide predictive maintenance alerts to ensure the vehicle remains in good condition.

3.2.6 System Integration and Workflow

Startup Mode: When the vehicle starts, the system will default to electric mode (using battery power). **Speed-based Mode Switching:** As the vehicle accelerates, the system will intelligently switch to LPG mode for speeds between 20-50 km/h and then to petrol mode for speeds above 50 km/h.

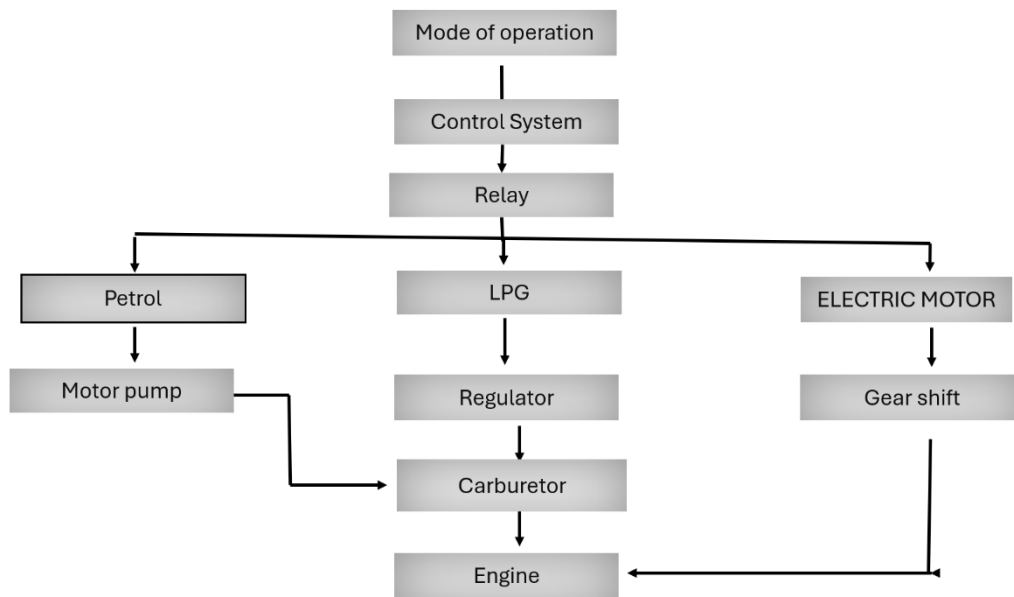


Figure 3.5 Mode of operation Flowchart

This ensures that the most appropriate power source is used for each driving condition. **Self-Recharging:** The alternator will charge the battery as the vehicle is running, reducing the need for external recharging. **Solar panels** will supplement the power during daylight hours. **Smart Monitoring and Control:** The ESP8266-based controller will ensure seamless switching between modes, monitor energy consumption, track the battery health, and manage system.

3.3 Flow chart Explanation

This flowchart represents the powertrain control and fuel source selection system for a tri-brid vehicle, which can run on petrol, LPG, or an electric motor. The process starts with the Mode of Operation, which is selected by the user or an onboard decision-making unit. Once the mode is selected, it is processed by the Control System, which then signals a Relay to activate the desired power source. The relay acts as a switch that routes the control to one of the three available propulsion systems. If Petrol mode is selected, the relay activates the Petrol path, which engages a Motor Pump to deliver fuel to the engine. If LPG (Liquefied Petroleum Gas) mode is selected, the relay directs power to the LPG system. Here, the gas first passes through a Regulator (which controls gas pressure), and then into a Carburetor, where it is mixed with air and sent into the Engine for combustion. If Electric Motor mode is selected, the control signal activates the Electric Motor, which is connected to the vehicle's drivetrain via a Gear Shift mechanism. The gear shift ensures proper torque and speed transfer to the wheels, similar to a transmission system in a conventional vehicle. Both the Engine (in the case of Petrol or LPG) and the Electric Motor can independently drive the vehicle based on the selected mode. This setup allows the vehicle to switch flexibly between internal combustion (petrol/LPG) and electric power, optimizing fuel efficiency, emissions, and performance based on driving conditions.

3.3.1 Power Sources and Charging

SystemSolar Panel:

Captures sunlight and converts it into electrical energy. Connected to a charge Controller to regulate the output voltage and current, preventing overcharging of the Lithium Battery.

Alternator:

Typically used in vehicles to generate electrical energy from the engine's mechanical power. Works alongside the Solar Panel as an additional charging source. Its output is also routed through the Charge Controller to ensure compatibility with the Lithium Battery.

AC Mains:

Serves as an external charging source for times when solar or alternator power is insufficient. Power from the AC mains is converted to DC using an AC to DC Converter. The output is passed through a Charge Controller before being stored in the LiFePO₄ Battery.

Lithium Battery:

Stores energy generated by the solar panel and alternator. It is lightweight and has a high energy density, making it suitable for Tri Brid vehicles. LiFePO₄ Battery (Lithium Iron Phosphate): Stores energy from the AC mains. It is highly durable, thermally stable, and provides additional storage for longer usage. A Switch BetweenBattery toggles between the two batteries based on the availability of charge or power demand.

3.3.2 Mode Selection and Power Transmission

Petrol Engine:

Provides propulsion when the vehicle operates in Petrol Mode. Useful for higher speeds or longer distances where battery-based or LPG power may not be sufficient.

LPG System:

Acts as a secondary fuel system. Provides an alternative to petrol, making the vehicle more economical and environmentally friendly. Works in combination with the Mode Shifting System.

Mode Shifting System:

Automatically selects the appropriate mode (Electric, Petrol, or LPG) based on vehicle speed or user input:

3.3.3.1 Below 20 km/h: Electric Mode (using batteries).

3.3.3.2 Between 20-50 km/h: LPG Mode.

3.3.3.3 Above 50 km/h: Petrol Mode.

3.3.3.4 Ensures a seamless transition between different modes for optimal performance and fuel efficiency.

3.3.3 Control and Propulsion System

Forward or Reverse Switch:

System Forward or Reverse Switch:

Provides directional control to the vehicle, allowing the user to select forward or reverse motion. This is critical for vehicle manoeuvrability, particularly in electric and Tri Brid systems.

Throttle Input:

Captures user commands to control the speed of the vehicle. The input is sent to the Motor Drive, which adjusts the power supplied to the motor.

Motor Drive:

Converts electrical energy from the batteries into mechanical energy. Regulates the power supplied to the Main Motor, ensuring smooth acceleration and deceleration. Works with the Throttle Input and Forward/Reverse Switch to maintain precise control over the vehicle's motion.

Main Motor:

1. Drives the wheels of the vehicle.
2. Receives power from the Motor Drive in electric mode, or indirectly through the Petrol Engine/LPG System in their respective modes.
3. **Display:**
 1. Provides real-time feedback to the user.
 2. Displays key information such as:
 3. Current mode (Electric, Petrol, or LPG).
 4. Battery charge levels for both Lithium and LiFePO4 batteries.
 5. Speed and power consumption.
 6. Fault or error notifications for troubleshooting.

3.3.4 Working Process**Power Generation and Storage:**

1. Solar panels and the alternator work together to charge the

Lithium Battery.

2. AC mains power is converted and stored in the LiFePO₄ Battery as a backup.

Mode Selection and Switching:

The Mode Shifting System determines whether the vehicle operates on Electric, LPG, or Petrol mode based on predefined speed thresholds.

Power Transmission to Motor:

Depending on the selected mode, power is routed from the appropriate source (batteries, petrol engine, or LPG system) to the Motor Drive.

User Controls:

The Throttle Input and Forward/Reverse Switch allow the user to control the speed and direction of the vehicle. Feedback is displayed on the Display Unit, ensuring the user stays informed about the system's status.

Tri Brid Efficiency:

The system ensures efficient use of available energy sources (solar, petrol, and LPG) while maximizing battery life and minimizing fuel consumption.

3.4 ADVANTAGES

- **Flexibility:** Multiple power sources provide greater adaptability to different scenarios.
- **Efficiency:** Automatic mode shifting ensures optimal energy use.
- **Sustainability:** Solar charging reduces dependence on non-renewable energy.

CHAPTER-4

TESTING OF PRODUCT

4.1 Testing of Product

4.1.1 Solar Panel Efficiency Test

Objective:

To measure the solar panels' energy conversion efficiency and ensure adequate power generation.

Procedure:

Place the vehicle in direct sunlight. Measure the power output of the solar panels using a multimeter or power analyser. Compare the generated power with the expected output based on the panel specifications and environmental conditions.

Expected Output:

Solar panels should achieve an efficiency of 15–20% under standard test conditions. Generated power should charge the battery at a stable rate.

4.1.2 Battery Charging and Discharging Test

Objective: Procedure:

To evaluate the battery's charging speed and energy retention capabilities. Charge the battery using solar panels, alternator (self-recharge system), and an external charging station. Measure the charging time and monitor the battery's charge retention under load conditions.

Expected Output:

The battery should reach full charge within the expected time. Battery charge should sustain the vehicle for the calculated range in electric mode.

4.1.3 Tri Brid Mode Switching Test**Objective:**

To ensure seamless switching between power modes (Electric, LPG, Petrol).

Procedure:

Operate the vehicle at varying speeds (e.g., below 20 km/h/between 20-50 km/h, above 50 km/h). Monitor the system's response to speed changes and verify the automatic switching between modes.

Expected Output:

Smooth and seamless transition between power modes without noticeable delays. Optimal mode selected for speed and driving conditions.

4.1.4 Regenerative Braking and Self-Recharge Test**Objective:**

To test the alternator and regenerative braking system for battery recharging.

Procedure:

Drive the vehicle in a controlled environment. Monitor the power generation from the alternator during motion and braking. Measure the additional charge delivered to the battery.

Expected Output:

Alternator should provide sufficient power to recharge the battery during vehicle operation. Regenerative braking should effectively generate electricity during deceleration.

4.1.5 ESP8266 Control and IoT Functionality Test**Objective:**

To verify the control system's ability to monitor and manage vehicle functions.

Procedure:

Test the ESP8266's performance in switching power modes, monitoring battery health, and displaying data. Connect the ESP8266 to the IoT system and test live GPS tracking and mobile app monitoring.

Expected Output:

Real-time data on battery status, power mode, speed, and GPS tracking should be displayed accurately. Smooth control of mode switching and reliable mobile app communication.

4.1.6 Emission Testing (LPG and Petrol Modes)**Objective:**

To ensure the Tri Brid system operates within permissible emission levels.

Procedure:

Test the vehicle in LPG and PETROL modes under varying load conditions. Use an emission analyser to measure carbon dioxide (CO₂), carbon monoxide (CO), and nitrogen oxides (Nox) emissions.

Expected Output:

Emissions should be significantly lower in LPG mode compared to petrolmode and within regulatory limits.

4.1.7. Road Testing**Objective:**

To assess the vehicle's overall performance on various terrains and conditions.

Procedure:

Drive the vehicle on city roads, highways and off-road conditions. Test acceleration, braking, power source switching, and system stability.

Expected Output:

Stable performance with smooth acceleration, efficient braking, and effective power management. Confirm the range for electric, LPG, and petrol modes.

4.1.8 Conclusion from Testing

The Solar Power Car with Tri Brid Petrol, LPG, Electric, by Self-Recharge System demonstrates excellent performance in energy efficiency, sustainability, and usability. It is a viable solution for reducing environmental impact and enhancing energy independence in modern transportation system.



Figure 4.1 Frame Work

CHAPTER-5

MODULES

5.1 Module Description

5.1.1 Power System Module

- **Solar Panel Setup:**
 - Converts sunlight into electricity.
 - Panels are arranged at optimal angles to maximize energy capture.
 - Energy is stored in the battery for later use.
 - Includes MPPT (Maximum Power Point Tracking) to optimize power output.



Figure 5.1 Solar Panel Set

- **Battery Management System (BMS):**
 - Manages charging and discharging cycles of the battery.
 - Monitors battery temperature, voltage, and current for safety and efficiency.

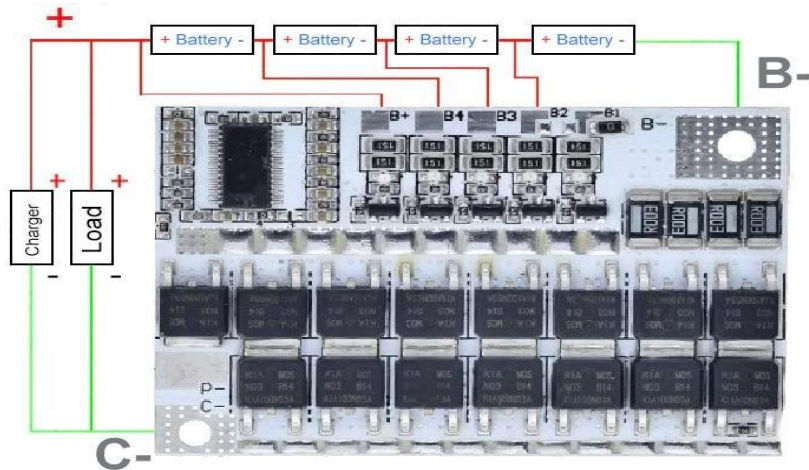


Figure 5.2 BMS

- **Self-Recharge Mechanism:**
 - Uses a dynamo or generator to generate power during vehicle operation.
 - Power is fed back into the battery to increase the range.



Figure 5.3 Self-Recharge Mechanis

5.1.2 Tri Brid Powertrain Module:

This module controls the vehicle's propulsion and mode switching.

Electric Drive:

- Includes an electric motor powered by the battery.
- Used for low-speed operation (<20 km/h) to ensure quiet and efficient performance.

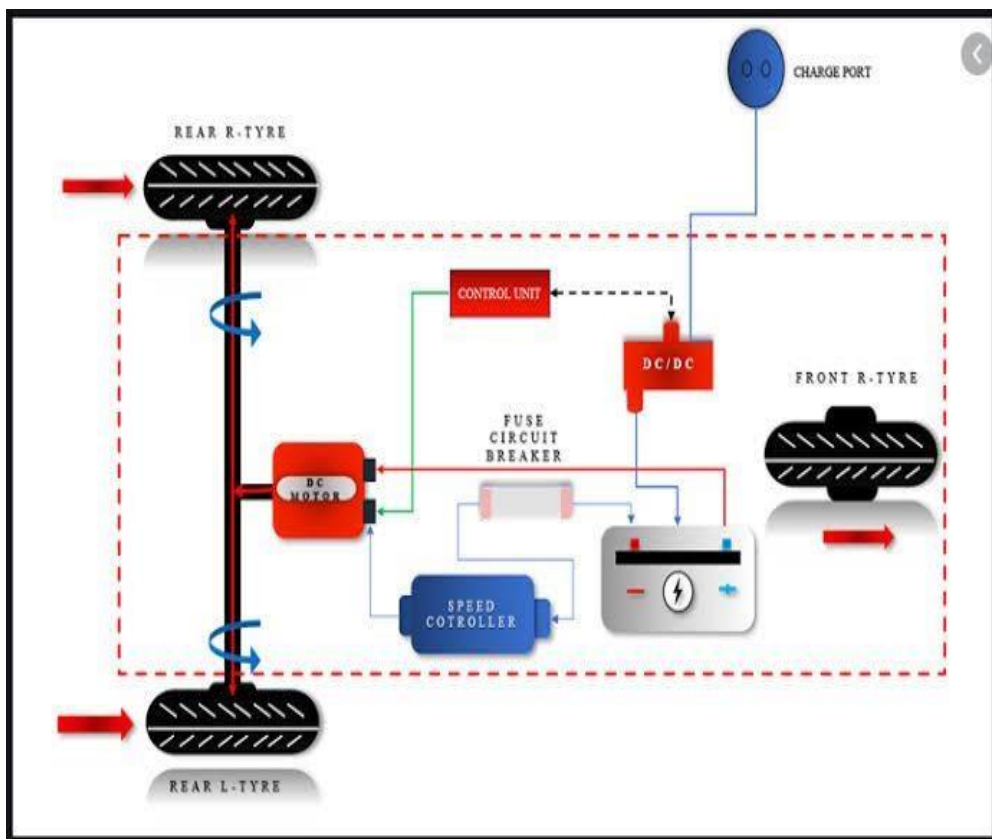


Figure 5.4 Electric Drive

LPG Engine Drive:

- Activates at medium speeds (20-50 km/h).
- Operates using compressed natural gas, reducing emissions and cost compared to petrol.

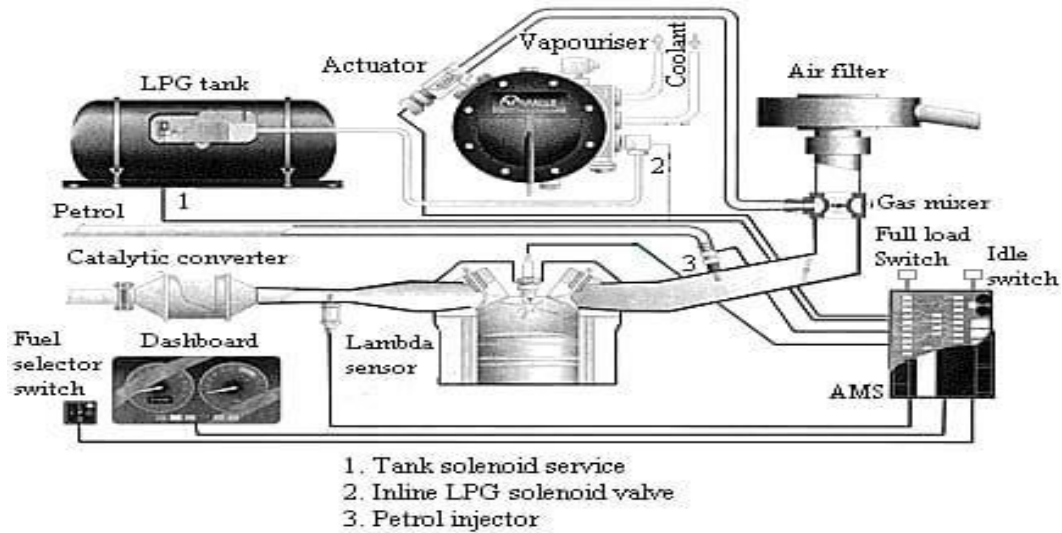


Figure 5.5 LPG Engine Drive

Petrol Engine Drive:

- Engages at high speeds (>50 km/h) for maximum power and torque.
- Ensures reliability for long-distance or high-speed travel.

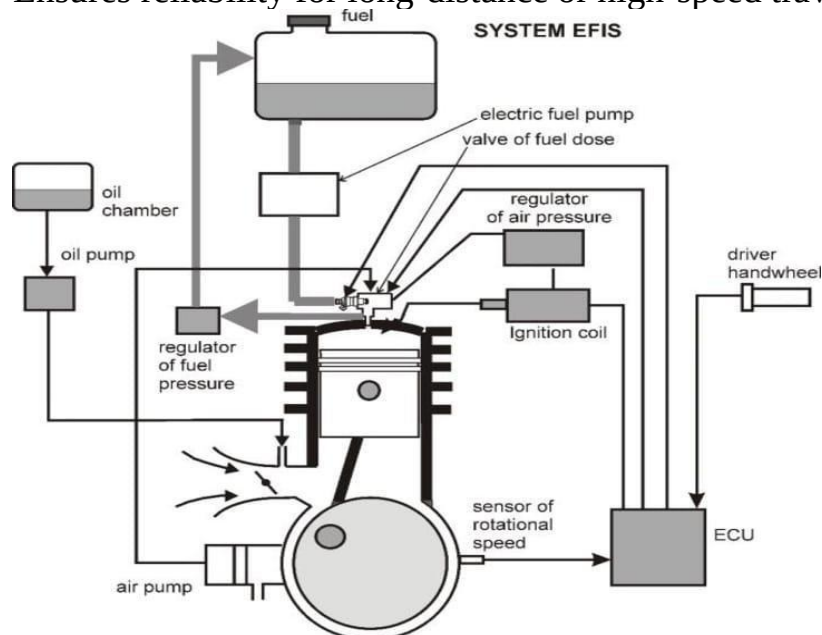


Figure 5.6 Petrol Engine Drive

Switching Mechanism:

- Automatic switching between modes based on speed thresholds.
- Integrated with sensors to monitor speed and control the transition smoothly.



Figure 5.7 Switching Mechanism

5.1.3. Control and Switching Module

Handles the overall control logic and switching between different operational modes.

- **Mode Selector System:**

- Uses speed sensors to determine the appropriate mode (electric, LPG, petrol).
- Ensures seamless transitions without driver intervention.

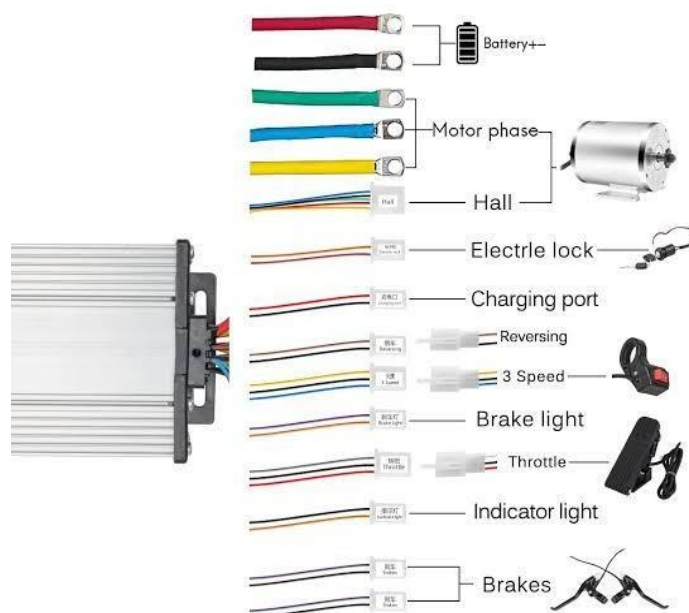


Figure 5.8 Control and Switching Module

- **IoT-Based Switch System:**

- Powered by ESP8266.
- Allows remote monitoring and control of vehicle mode via a mobile application.
- Provides alerts for mode changes, battery levels, and system faults.

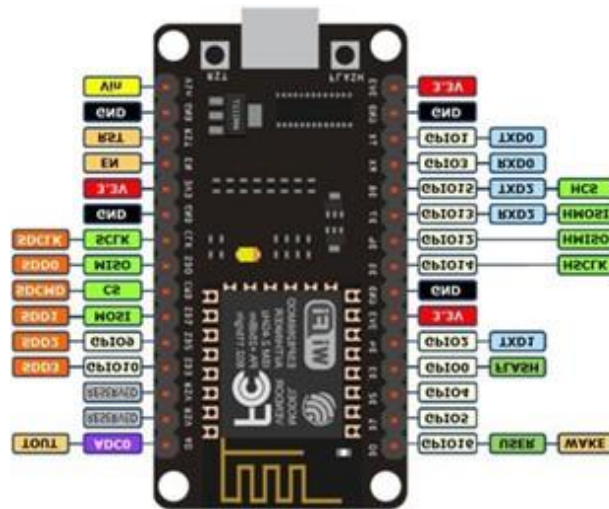


Figure 5.9 IoT-Based Switch System

5.1.4 Monitoring Module

This module focuses on real-time data acquisition and display.

GPS Live Map System:

- Tracks vehicle location and displays routes on a mobile interface.
- Can be used for navigation and geofencing purposes.



Figure 5.10 GPS

Live Map System

Live Data Monitoring:

- Monitors battery charge level, power generation, speed, and current mode.
- Displays information on a dashboard or mobile app for user convenience.



1

Figure 5.11 Live Data Monitoring

5.1.5 Safety and Testing Module

Ensures the vehicle operates safely and efficiently.

Safety Mechanisms:

1. Includes overload protection, short-circuit protection, and temperature control.
2. Emergency cut-off switches for critical situations.

Testing Procedures:

3. Tests performed under varying speeds, loads, and environmental conditions.
4. Verifies efficiency of self-recharge system and seamless mode transitions.

5.1.6 Conversion System Module

This module is crucial for integrating and switching between propulsion systems.

- **Engine-Electric Motor Integration:**

- Implements hardware and software controls to coordinate the engine and motor.
- Ensures both systems can operate independently and together as needed.

- **Power Electronics:**

- Uses inverters to convert DC power from the battery to AC for the electric motor.
- Converts AC from the dynamo to DC for battery charging.

5.1.7 Auxiliary Module

Adds additional functionalities and enhances the user experience.

- **User Interface Design:**

5.1.7.1 Develops an intuitive mobile app or dashboard for operation and monitoring.

5.1.7.2 Includes alerts for low battery, high temperature, and system status.

- **Lighting and Auxiliary Systems:**

5.1.7.3 Provides power to auxiliary components like lights, indicators, and horn using the battery.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 SOFTWARE REQUIREMENTS:

6.1.1 Development Platforms

- **Arduino IDE:**

- For programming microcontrollers ESP8266.

6.1.2 Monitoring and IoT

- **Blynk or Things Board:**

- Platforms for IoT-based live monitoring and control.

- **Google Maps API:**

- For GPS live mapping and routing.

6.2 HARDWARE REQUIREMENTS:

6.2.1. Hardware Requirements

Power System Components

Solar Panels:

Solar panels are devices that convert sunlight into electricity using the photovoltaic (PV) effect. Made primarily from semiconductor materials like silicon, solar panels are used in a wide range of applications—from residential rooftops to large-scale solar farms. As concerns about renewable energy and sustainability grow, solar technology is also being integrated into transportation, particularly in solar-powered cars. In these vehicles, solar panels are either installed on the car's surface or used to charge onboard batteries. While the amount of energy generated by these panels is usually not enough to power a car entirely, they can extend the driving range, reduce reliance on external charging, and support small systems like air conditioning

or lighting. Some experimental and concept vehicles, like those used in solar car races or developed by companies for research, aim to be fully solar-powered, but commercial use is still limited due to the low energy output and high cost. However, as solar technology continues to improve, its role in automotive design is expected to grow, helping to make transportation cleaner and more efficient



Figure 6.1 Solar Panels

Battery:

The battery used in many modern electric systems, including solar-powered vehicles and energy storage units, is typically a lithium-ion type due to its high efficiency and long lifespan. A commonly preferred specification for optimal performance is a 48V, 30Ah lithium-ion battery. This configuration offers a strong balance between power output and energy storage capacity, making it suitable for applications that require sustained energy delivery over time. The 48-volt system provides adequate voltage for most electric motors and systems, while the 30 ampere-hour (Ah) capacity ensures a decent runtime before needing to recharge. Lithium-ion batteries are also known for being lightweight, fast-charging, and low-maintenance compared to older battery technologies like lead-acid, making them a popular and practical choice in modern energy systems.



Figure 6.2 Battery (lithium Battery)

Battery Management System (BMS):

A **Battery Management System (BMS)** is a crucial component in any device or system that uses rechargeable batteries, especially lithium-ion batteries. Its primary function is to monitor and manage the performance, safety, and health of the battery pack. The BMS continuously checks vital parameters such as voltage, current, temperature, and the state of charge (SOC) of individual battery cells. It ensures that all cells are balanced meaning they charge and discharge evenly which helps maximize the battery's efficiency and lifespan. In addition to monitoring, the BMS protects the battery from critical issues like overcharging, over-discharging, overheating, or short-circuiting, all of which can damage the battery or even pose safety risks like fires. It may also provide communication between the battery and external systems, such as displaying battery status to the user or controlling when the battery should be charged or disconnected. Overall, a BMS is essential for safe and efficient operation, particularly in applications like electric vehicles, solar energy storage systems, and portable electronics.

Engine coil:

The ignition coil in a two-stroke auto rickshaw engine is a vital component of the ignition system, responsible for generating the high voltage needed to ignite the fuel-air mixture in the engine's cylinder. In a two-stroke engine, the coil transforms the low voltage from the battery or magneto into a much higher voltage, typically between 12,000 to 20,000 volts. This high voltage is then sent to the spark plug, creating a spark that ignites the compressed mixture, powering the engine. Unlike four-stroke engines, two-stroke engines fire once every revolution, which means the ignition coil must operate more frequently and reliably. The coil consists of two windings: a primary winding that

receives the low voltage and a secondary winding that produces the high voltage through electromagnetic induction. This system must be well-timed and efficient, as any delay or failure in producing the spark can cause misfires, rough idling, or difficulty starting the engine. Over time, issues such as coil overheating, internal wear, or electrical faults can affect performance, making regular inspection and maintenance essential for smooth operation in two-stroke auto rickshaws.



Figure 6.3 Dynamo coil

6.2.2 Propulsion System Components

- **Electric Motor:**

A **1000W DC electric motor** is a powerful motor commonly used in various electric vehicles, including electric bikes, scooters, and auto rickshaws, as well as in industrial and hobby applications. The "1000W" rating indicates that the motor can output 1000 watts of power, which is equivalent to approximately 1.34 horsepower. Being a **DC (Direct Current)** motor, it operates using a constant voltage source, typically from a battery pack, and is known for its smooth and efficient

performance. DC motors offer excellent torque at low speeds, making them ideal for vehicles that require quick acceleration and climbing ability. A 1000W DC motor generally operates on a voltage range between **48V to 60V**, depending on its design. These motors can reach speeds of up to **40–50 km/h**, depending on the load, gearing, and efficiency. One of the key advantages of DC motors is their **simplicity and ease of control**, especially with modern motor controllers that allow for variable speed, direction control, and braking. They are also relatively quiet and require less maintenance compared to combustion engines, as they have fewer moving parts. However, efficient cooling and a good controller system are essential to prevent overheating and ensure long-term performance. Overall, a 1000W DC motor is a reliable and efficient choice for medium-duty electric propulsion systems.



Figure 6.4 Electric Motor

LPG Engine:

The **LPG (Liquefied Petroleum Gas) regulator system** in a car engine is a vital component that ensures the safe and efficient use of LPG as an alternative fuel. This system is designed to control the pressure and flow of LPG from the storage tank to the engine, converting the liquid fuel into a vapor that can be properly mixed with air and burned in the combustion chamber. In LPG-powered cars, the fuel is stored in a pressurized tank in liquid form. When the engine is running, LPG is directed through a **regulator or vaporizer**, which performs two key functions: it **reduces the high pressure** of the fuel and **converts it from liquid to gas** (vapor). This vapor is then delivered to the engine through a mixer or injector system, depending on whether the vehicle uses a carbureted or fuel-injected setup. The LPG regulator works in conjunction with **solenoid valves** and **electronic control units (ECUs)** to ensure optimal fuel delivery. Solenoid valves open and close automatically to allow gas flow only when the engine is on, providing an important safety feature. The ECU adjusts the fuel-air mixture based on engine conditions such as speed, temperature, and load, ensuring efficient combustion and reduced emissions.



Figure 6.5 LPG Engine

Petrol Engine:

A **2-stroke petrol engine** is a type of internal combustion engine that completes a power cycle in just two strokes of the piston (one revolution of the crankshaft), unlike the 4-stroke engine which requires four strokes. In a 2-stroke petrol engine, the intake, compression, combustion, and exhaust processes are combined into just two piston movements, making it simpler and lighter. When the piston moves up, it compresses the air-fuel mixture while simultaneously creating a vacuum in the crankcase to draw in a fresh charge. As the piston moves down after ignition, it exposes ports that allow the exhaust gases to exit and the fresh mixture to enter. These engines are commonly used in motorcycles, scooters, chainsaws, and small power tools due to their high power-to-weight ratio and mechanical simplicity, though they are less fuel-efficient and more polluting than 4-stroke engines.

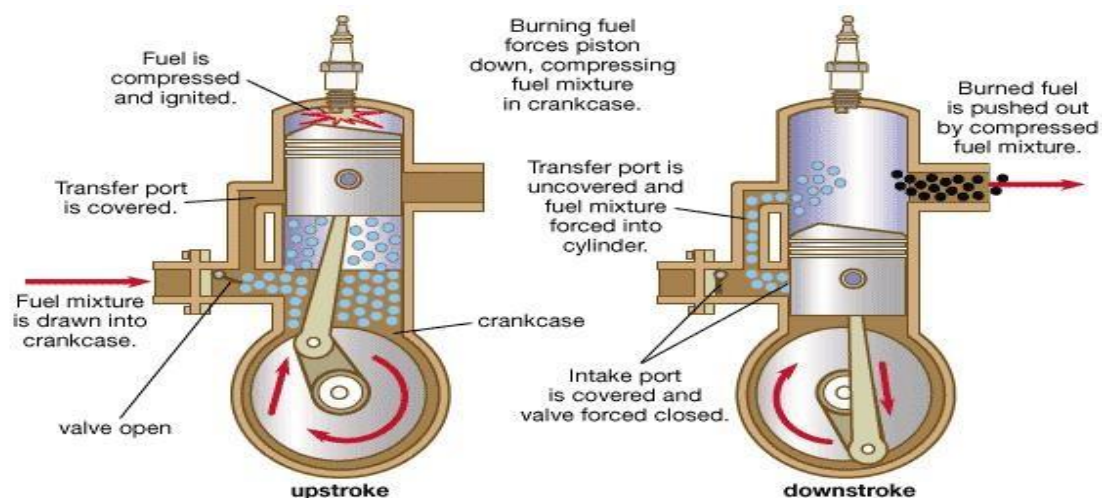


Figure 6.6 Petrol Engine

Controller:

An **EV controller for a 1000W motor** is a crucial component in electric vehicles such as e-bikes, scooters, and small electric motorcycles. It serves as the interface between the battery and the motor, regulating the flow of electrical energy to ensure smooth and efficient operation. Typically designed for use with brushless DC (BLDC) motors, the controller converts the battery's direct current (DC) into a form suitable for powering the motor, while also managing functions such as acceleration, braking, and speed control. A 1000W controller is usually compatible with 24V, 36V, or 48V battery systems, depending on the motor specifications. It receives input from the throttle and responds accordingly by adjusting power output. Many models include safety features like over-voltage, under-voltage, over-current, and thermal protection to prevent damage to the motor and battery. Some advanced controllers may also support regenerative braking and offer connectivity for LCD displays or Bluetooth modules. Overall, the EV controller plays a vital role in the performance and safety of an electric vehicle.

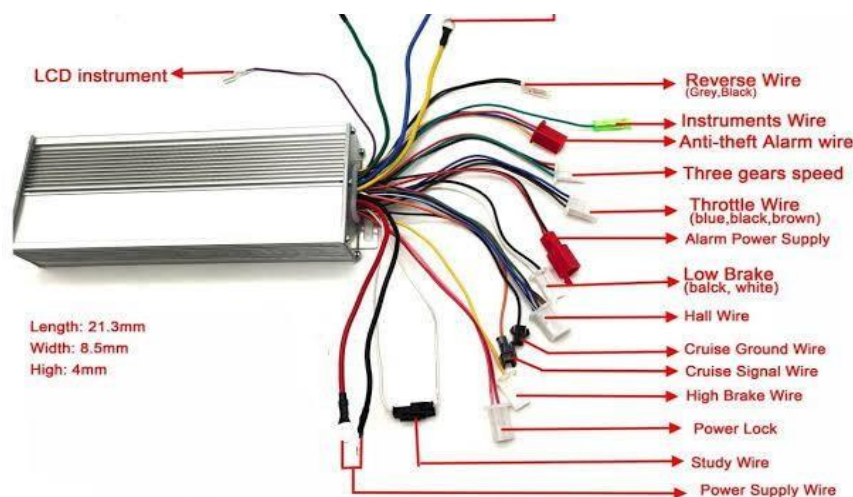


Figure 6.7 Controller

6.2.3. Conversion System Components

Converters:

Converters in Electric Vehicles (EVs) play a vital role in managing the flow and conversion of electrical energy for efficient motor operation and battery management. Two important types of converters used are the **DC-to-AC inverter** and the **AC-to-DC converter**. The **DC-to-AC inverter** is responsible for converting the direct current (DC) from the vehicle's battery into alternating current (AC) required by the motor, especially in vehicles that use AC motors. This conversion is essential because many high-efficiency EV motors operate on AC power, which allows for better performance, smoother speed control, and regenerative braking capabilities. On the other hand, the **AC-to-DC converter** is used when charging the battery from an AC source, such as a dynamo or generator. This converter transforms the alternating current generated during motion or braking into direct current, which is suitable for recharging the EV's battery. This process is especially useful in systems with regenerative braking, where the vehicle's kinetic energy is converted into electrical energy and stored back in the battery, improving overall efficiency and range. Together, these converters ensure that power is effectively managed and transferred within the vehicle, optimizing both driving performance and energy recovery.

The switching mechanism in an electric vehicle (EV) or hybrid system is crucial for managing and automating the transition between different operating modes, such as motor drive, regenerative braking, and battery charging. This mechanism often relies on a combination of relay modules and microcontroller-based systems to perform seamless switching operations. Relay modules act as electrically operated switches that control high-current circuits using low-power signals. They are used to direct power flow between components like the motor, battery, and charger. However, relays alone are not intelligent they need to be controlled. This is where microcontroller-based systems come in. Microcontrollers monitor various inputs such as speed, throttle position, battery status, and braking signals. Based on predefined logic or conditions, the microcontroller sends commands to the relay modules to switch modes automatically. For example, it can switch from driving mode to charging mode when regenerative braking is detected, or from battery power to external charging when plugged into a power source. Together, the relay modules and microcontroller ensure a smart, efficient, and safe operation of the EV by automating the switching process without requiring manual intervention from the user.



6.2.4 IoT and Monitoring Components

Microcontroller:

The **ESP8266** is a low-cost, Wi-Fi-enabled microcontroller that is widely used in electronics and IoT (Internet of Things) applications, and it can also play an important role in electric vehicle (EV) systems. Developed by Espressif Systems, the ESP8266 combines a microcontroller unit (MCU) with built-in Wi-Fi capabilities, making it ideal for wireless communication and remote monitoring tasks. In the context of EVs, the **ESP8266 microcontroller** can be used to control and automate various functions, such as switching between operational modes (drive, charge, regenerative braking), monitoring battery levels, and communicating system status to a mobile app or cloud platform. Its GPIO (General Purpose Input/Output) pins can interface with sensors, relay modules, and other control circuits. With support for programming via the Arduino IDE and integration with platforms like Blynk or MQTT, the ESP8266 allows for real-time data logging, remote diagnostics, and even wireless updates. Overall, the ESP8266 adds intelligence and connectivity to an EV system, enabling smarter control, enhanced user interaction, and integration with modern IoT infrastructure.

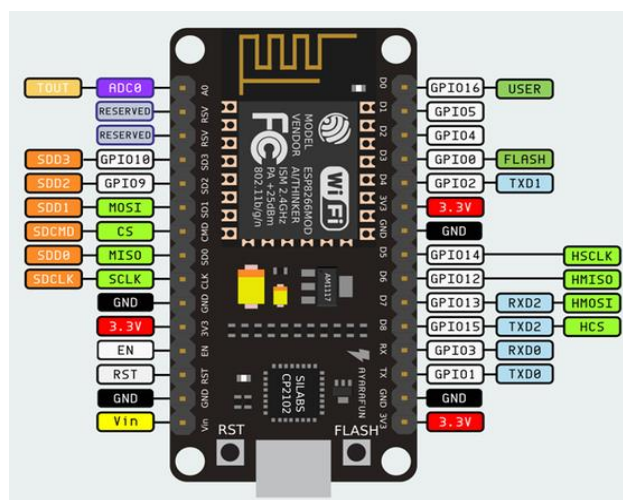


Figure 6.9 Microcontroller

Sensors:

The **DHT11 sensor** is a low-cost, basic digital sensor used for measuring **temperature and humidity**. It is widely used in DIY electronics, weather monitoring systems, and IoT applications due to its simplicity and ease of integration with microcontrollers like Arduino, ESP8266, and Raspberry Pi. The DHT11 uses a **thermistor** for temperature measurement and a **capacitive humidity sensor** to detect the surrounding air's moisture level. It provides temperature readings typically in the range of **0°C to 50°C** with an accuracy of $\pm 2^\circ\text{C}$, and humidity readings from **20% to 90% RH** with an accuracy of $\pm 5\%$. The sensor outputs data in digital form, making it easy to read via a single data pin, which reduces the need for complex analog-to-digital conversions. Though it has a relatively slow sampling rate (about one reading per second) and limited accuracy compared to more advanced sensors like the DHT22 or BME280, the DHT11 is a popular choice for basic environmental monitoring due to its affordability and ease of use.

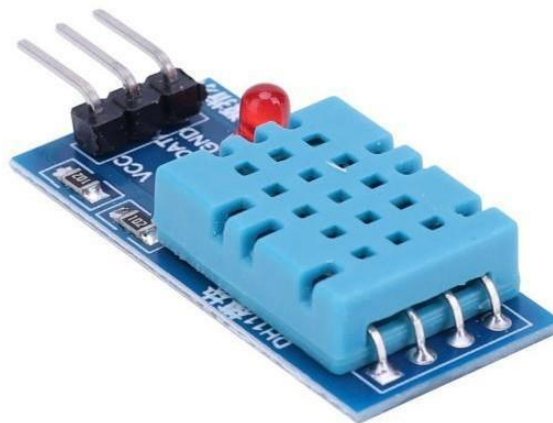


Figure 6.10 Sensors

GPS Module:

A **real-time GPS module** is a device used to determine the precise location of an object, vehicle, or person by receiving signals from **Global Positioning System (GPS)** satellites. These modules are widely used in navigation systems, vehicle tracking, asset monitoring, and various Internet of Things (IoT) applications. A GPS module typically includes a **GPS receiver**, a **microcontroller interface**, and often a built-in antenna. It communicates with satellites to calculate its exact position in terms of **latitude**, **longitude**, **altitude**, and **time**. Some modules also provide speed and heading information. Popular GPS modules include the **Neo-6M**, **Neo-7M**, and **Neo-M8N**, which are known for their reliability and ease of integration with microcontrollers like Arduino and ESP8266. In real-time applications, the GPS module continuously updates the location data, which can then be transmitted to a central server or mobile app using communication technologies such as GSM, Wi-Fi, or Bluetooth. This enables **live tracking** of vehicles or assets. The module usually communicates with the host system via **UART (serial)** protocol, outputting standard **NMEA sentences**, which are easy to parse and use for location-based functions. Overall, a real-time GPS module plays a vital role in enabling **live navigation**, **tracking**, and **geo-fencing** features in modern embedded systems and smart devices.

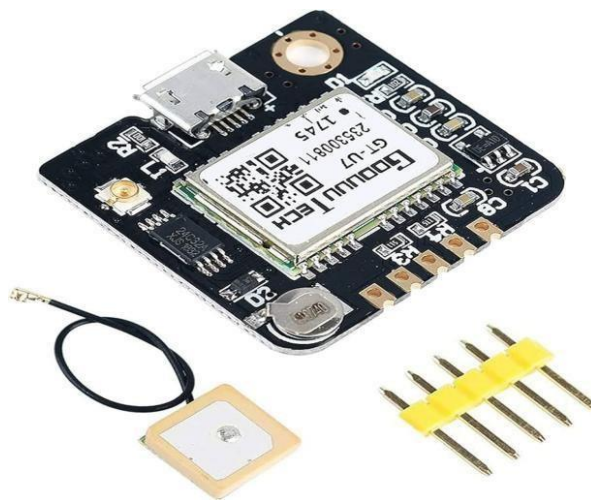


Figure 6.11 GPS Module

Display System:

The **Android Display System** refers to the visual interface used in Android devices, which includes smartphones, tablets, and embedded systems running the Android operating system. This system manages how information is presented on the screen, allowing users to interact with apps, view data, and control connected hardware. In the context of embedded or IoT projects, an Android Display System often involves using an Android-powered device or tablet as a user interface, showing real-time data such as sensor readings, GPS navigation, battery status, and other important system information. The display can be a simple LCD screen or a more advanced touchscreen panel, enabling users not only to see information but also to interact through taps, swipes, and gestures. Thanks to Android's flexible software ecosystem, developers can create customized apps that communicate with hardware components via Bluetooth, Wi-Fi, or USB, providing a rich and intuitive user experience. Overall, the Android Display System serves as a powerful and versatile platform for monitoring, controlling, and interacting with smart devices and systems.



Figure 6.12 Display System

6.2.5 Safety and Auxiliary Components

Circuit Breakers and Fuses:

Circuit breakers and fuses are crucial components in electrical systems designed to protect circuits from damage caused by overcurrent, short circuits, or electrical faults. Both devices serve the primary purpose of interrupting the flow of excessive electrical current to prevent overheating, fire hazards, and damage to electrical components, but they differ in design and operation. A fuse is a simple and cost-effective protective device that contains a thin metal wire or strip which melts when the current exceeds a specified threshold, thereby breaking the circuit and stopping the flow of electricity. Once a fuse blows, it must be replaced, making it a one-time-use safety device. Fuses are widely used due to their reliability and simplicity but can be inconvenient when frequent replacements are needed. On the other hand, a circuit breaker is an automatic switch that detects excess current or faults and trips to cut off the electrical supply. Unlike fuses, circuit breakers can be reset either manually or automatically after tripping, eliminating the need for replacement and allowing for quicker restoration of power. Circuit breakers also offer additional features such as adjustable trip settings and faster response times. Both devices are essential for maintaining electrical safety in various applications, including residential, commercial, automotive, and industrial systems, helping to safeguard equipment, prevent accidents, and ensure smooth and safe operation of electrical circuits.



Figure 6.14 Circuit Breakers and Fuses

6.3 SOFTWARE DESCRIPTION:

The software for the Solar Power Car with Tri Brid Petrol, LPG, Electric, by Self-Recharge System is designed to integrate control, monitoring, and communication functionalities seamlessly. It utilizes embedded programming to manage the various operational modes of the Tri Brid system. The Arduino microcontroller, programmed via the Arduino IDE, handles real-time sensor data collection and controls essential operations such as speed monitoring, mode switching, and power flow management. For advanced processing and IoT integration, an ESP8266 programmed in Python acts as the central hub, enabling data logging, remote control, and live monitoring. The system incorporates IoT functionality using platforms like Blynk or Firebase, allowing users to monitor critical parameters such as battery status, speed, and operational mode through a mobile application. This app also facilitates remote mode switching and provides notifications in case of system faults or critical battery levels. A GPS module, integrated with the ESP8266, uses the Google Maps API to deliver real-time location tracking and routing, enhancing the system's usability. To ensure efficiency and safety, the software employs a fault detection mechanism that monitors for issues like overheating or overcharging and triggers an emergency cut-off if required. Additionally, a user-friendly interface on the mobile app or dashboard displays real-time energy usage, operational status, and diagnostic information, ensuring the Tri Brid system operates efficiently and reliably.

6.4 HARDWARE DESCRIPTION

The hardware components for the Solar Power Car with Tri Brid Petrol,LPG, Electric, by Self-Recharge System are designed to ensure efficient energy utilization, seamless mode transitions, and real-time monitoring. The primary hardware includes solar panels for energy generation, connected to a battery management system (BMS) that regulates the charging and discharging of high-capacity lithium-ion batteries. A combination of propulsion systems, including an electric motor, a CNG engine, and a petrolengine, is integrated to allow operation in different modes based on speed requirements.

To enable self-recharging, the system incorporates a dynamo or regenerative braking mechanism that generates electricity during motion. The hardware is controlled by a ESP8266 or microcontroller, which manages IoT-based functionalities, including remote monitoring and mode switching. Additional components such as speed sensors, inverters, and converters ensure proper energy flow between the battery and the motor. For real-time location tracking and monitoring, a GPS module and a mobile app interface are used. The vehicle is also equipped with safety features like temperature sensors, overload protection, and emergency cut-off switches for reliable operation. Auxiliary hardware, including LED lights and indicators, are powered by the battery for enhanced user experience

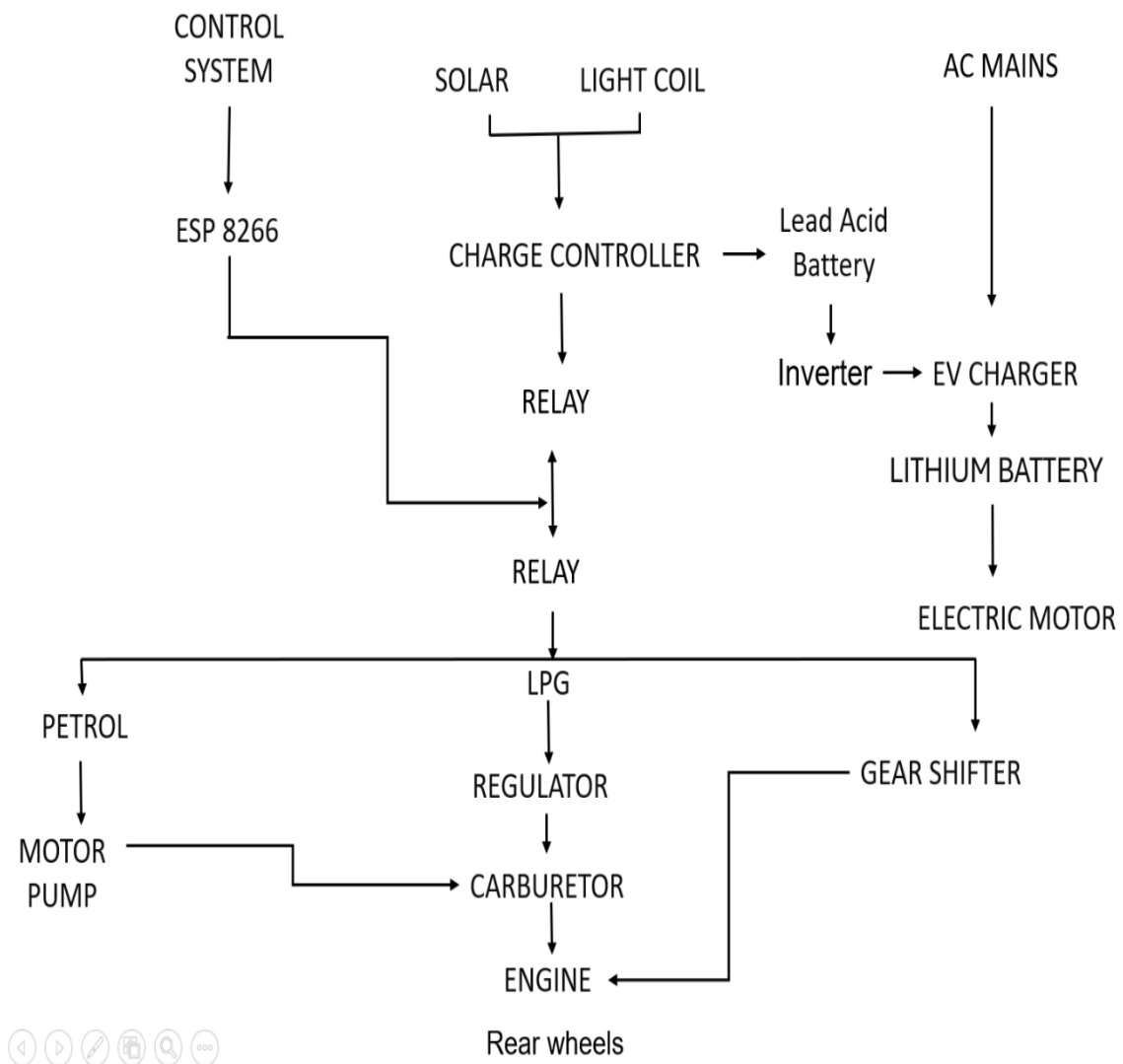


Figure 6.15 Flowchart of Control system

CHAPTER 7

RESULTS

7.1 Successful Integration of Tri Brid Power Sources:

- The vehicle operates seamlessly across three modes: electric, LPG, and petrol, with automatic switching based on speed thresholds.
- The solar panels provide a consistent source of energy, reducing reliance on external charging infrastructure.

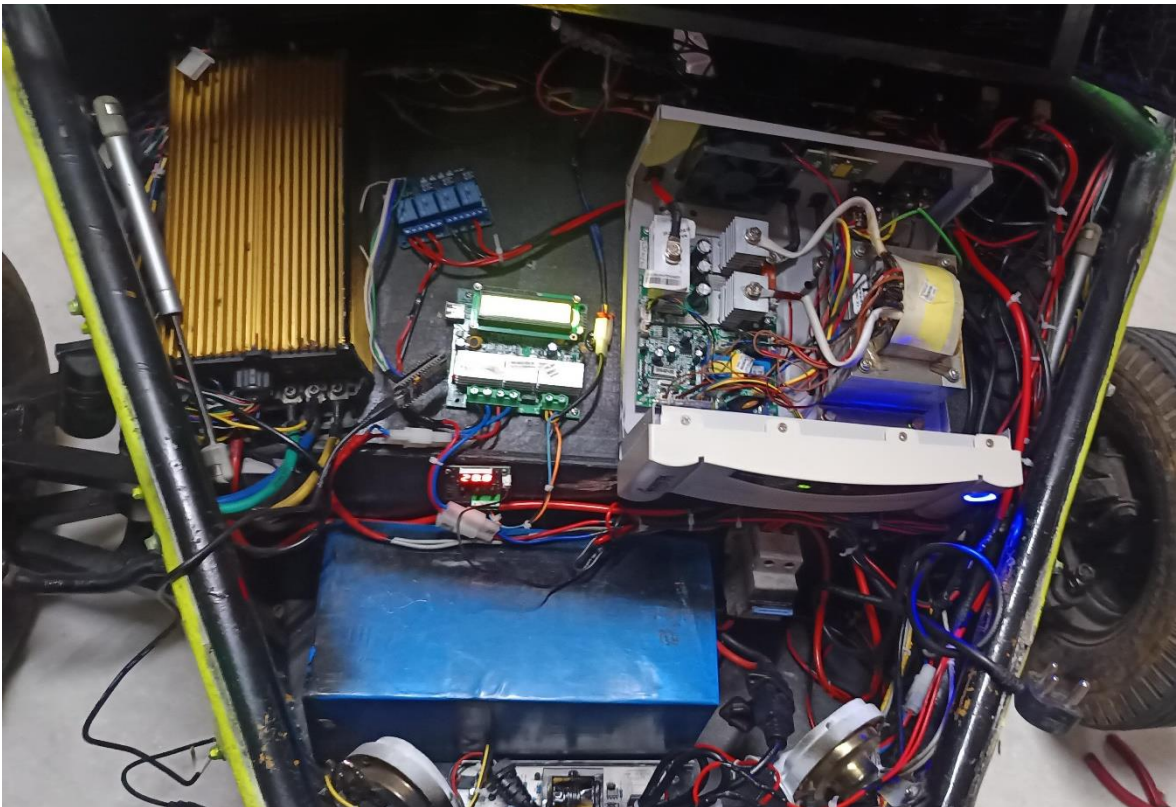


Figure 7.1 Ev wiring system testing



7.2 Efficient Self-Recharge Mechanism:

- The implemented dynamo system effectively recharges the battery during vehicle operation, increasing the range of the electric mode.

7.3 Enhanced Monitoring and Control:

- The IoT-based system using ESP8266 enables remote monitoring and control through a mobile application, including real-time GPS tracking and system diagnostics.

7.4 Environmental Benefits:

- The Tri Brid system significantly reduces emissions by utilizing solar energy and CNG, promoting cleaner transportation.

7.5 User-Friendly Design:

- The integration of a live dashboard and GPS system provides users with an intuitive interface for tracking performance and vehicle status.

FUTURE IMPLEMENTATIONS:

1. Advanced Solar Panel Integration:

- Use lightweight, flexible solar panels that can conform to the shape of the vehicle's roof, hood, or doors.
- Explore solar paint technology that converts sunlight into energy through nano-coating materials.
- Implement solar tracking systems that adjust panel angles to maximize sunlight capture throughout the day.

2. AI-Based Energy Management System (EMS):

- Develop smart systems that automatically switch between solar, petrol, and LPG based on energy availability and driving needs.
- Use predictive analytics to optimize driving routes for maximum solar exposure and fuel efficiency.
- Incorporate machine learning to study and adapt to the user's driving patterns for better energy conservation.

3. Plug-In Charging Option:

- Add plug-in Tri Brid support to allow battery charging from conventional power grids, especially useful in low sunlight areas.

4. Regenerative Suspension System:

- Implement energy recovery from suspension movement over rough terrains, turning mechanical stress into electrical energy.

5. Modular Fuel and Battery Systems:

- Design quick-swap LPG and petrol tanks for efficient fueling.
- Develop battery swapping modules for commercial applications such as fleet vehicles.

6. Use of Lightweight Sustainable Materials:

- Utilize carbon fiber, graphene, or recycled plastics for the vehicle body and interior components to improve efficiency and reduce weight.

7. IoT and Mobile App Integration:

- Create a mobile application for real-time vehicle monitoring, solar charging status, fuel levels, and driving analytics.
- Add an eco-driving coach that provides efficiency tips during travel.

8. Vehicle-to-Grid (V2G) and Vehicle-to-Home (V2H) Capabilities:

- Enable the vehicle to act as a power source during emergencies by supplying energy back to the grid or a household system.

9. Integration with Smart City Infrastructure:

- Connect the vehicle to smart traffic systems for real-time navigation, fuel-saving routes, and automatic toll or parking payments.

10. Autonomous Solar Parking:

- Develop an auto-parking system that detects and selects sunny parking areas to maximize solar recharging when the vehicle is idle.

11. Climate-Adaptive Ventilation:

- Use solar energy to power interior cooling and air circulation systems that adapt to external and internal temperature changes.\

12. Dual-Mode Onboard Generator:

- Include a compact generator capable of running on either petrol or LPG to charge the battery during low sunlight conditions.

13. Government and Public Sector Integration:

- Propose partnerships with local governments for using Tri Brid vehicles in municipal fleets to reduce city emissions.

14. Expandable Solar Surfaces:

- Design rooftop extensions or solar trailers to increase the surface area available for solar charging during long journeys or stationary periods.

15. Emergency Backup System:

- Enable the vehicle to serve as an emergency energy hub during power outages or natural disasters, powering essential household appliances or devices

16. Safety Features Powered by Solar:

- Utilize solar power for key safety systems including 360-degree cameras, proximity sensors, and automatic braking mechanisms.

17. Adaptive Aerodynamic Design:

- Use movable parts that adjust to driving conditions to reduce drag and increase overall energy efficiency.

18. Blockchain for Energy Transparency:

- Implement blockchain to securely track and record the energy source (solar, petrol, LPG) used, allowing for carbon credit validation and smart energy incentives.

19. Remote Diagnostics and Predictive Maintenance:

- Use embedded sensors and cloud-based diagnostics to predict and prevent breakdowns through timely alerts and maintenance notifications.

20. In-Vehicle Training Module:

- Introduce a built-in educational system that trains drivers on how to maximize energy efficiency and understand the working of the Tri Brid system.

Conclusion:

The development of a Self-Recharging Solar Tri Brid Vehicle with Integrated Petrol and LPG Systems represents a significant step toward sustainable and flexible transportation. By combining solar energy with dual-fuel internal combustion (petrol and LPG), this vehicle leverages the strengths of each energy source to reduce environmental impact, enhance fuel efficiency, and increase operational reliability.

The integration of a self-recharging solar system enables partial energy independence and supports reduced dependence on fossil fuels, especially for short-distance or urban commuting. Meanwhile, the inclusion of both petrol and LPG systems ensures that the vehicle remains functional across various terrains and fuel availability scenarios, making it ideal for both urban and remote applications.

This Tri Brid approach offers:

- Reduced emissions, thanks to cleaner LPG combustion and solar electric operation.
- Improved energy efficiency, through intelligent energy management and regenerative charging.
- Operational flexibility, with multiple fuelling options ensuring continued usability even in energy-scarce areas.

In conclusion, this project demonstrates the feasibility and potential impact of a multi-source Tri Brid vehicle that aligns with current environmental and energy goals. With further research, optimization, and investment, this concept could serve as a model for the next generation of eco-friendly, high-efficiency transportation systems.

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SOURCE CODE

```
#define BLYNK_TEMPLATE_ID "TMPL3TIAAndECa"
#define BLYNK_TEMPLATE_NAME "Bike control"
#define BLYNK_AUTH_TOKEN "HMRnDuld9LSMUie--mlj7FMS_KDN2Chb"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <DHT.h>
#include <BlynkSimpleEsp8266.h>
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "Logesh";
char pass[] = "logi2612";
DHT dht(D3, DHT11); //(sensor pin,sensor type)
BlynkTimer timer;
void DHT11sensor() {
    float h = dht.readHumidity();
    float t = dht.readTemperature();

    if (isnan(h) || isnan(t)) {
        Serial.println("Failed to read from DHT sensor!");
    }else if (t>=60){
        digitalWrite(D5,LOW);
    }
    Blynk.virtualWrite(V0, t);
    Blynk.virtualWrite(V1, h);
}
BLYNK_WRITE(V2)
{
    int value = param.asInt();
    Serial.println(value);
    if(value == 1)
    {
```

```
    digitalWrite(D5, HIGH);  
    Serial.println("LED ON");  
}  
if(value == 0)  
{  
    digitalWrite(D5, LOW);  
    Serial.println("LED OFF");  
}  
}
```

```
BLYNK_WRITE(V3)  
{  
    int value = param.asInt();  
    Serial.println(value);  
    if(value == 1)  
    {  
        digitalWrite(D6, HIGH);  
        Serial.println("LED ON");  
    }  
    if(value == 0)  
    {  
        digitalWrite(D6, LOW);  
        Serial.println("LED OFF");  
    }  
}
```

```
BLYNK_WRITE(V4)  
{  
    int value = param.asInt();  
    Serial.println(value);  
}
```

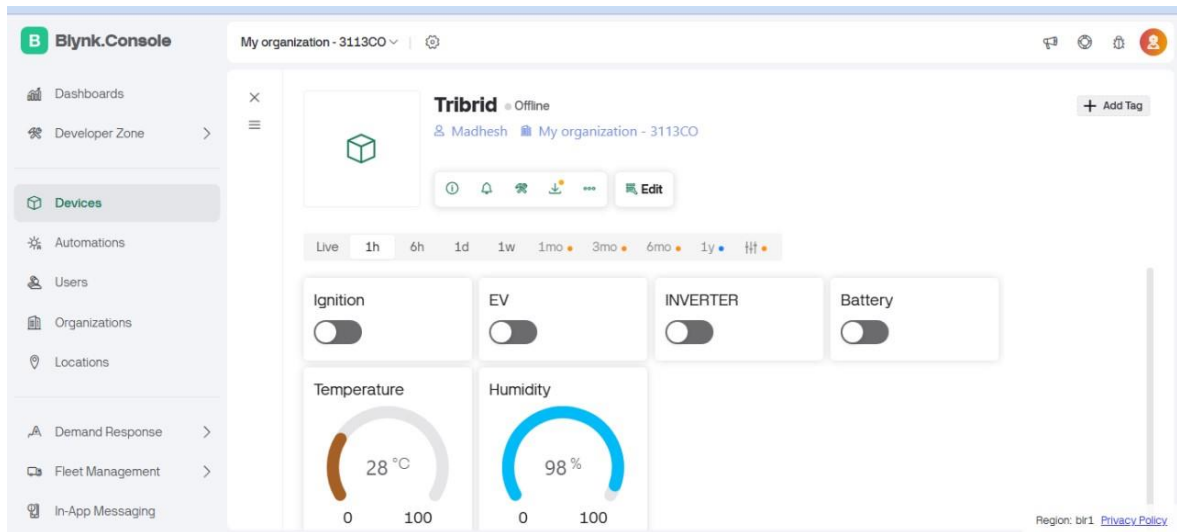
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if(value == 1)
{
    digitalWrite(D7, HIGH);
    Serial.println("LED ON");
}
if(value == 0)
{
    digitalWrite(D7, LOW);
    Serial.println("LED OFF");
}
}
```

```
BLYNK_WRITE(V5)
{
    int value = param.asInt();
    Serial.println(value);
    if(value == 1)
    {
        digitalWrite(D8, HIGH);
        Serial.println("LED ON");
    }
    if(value == 0)
    {
        digitalWrite(D8, LOW);
        Serial.println("LED OFF");
    }
}
```

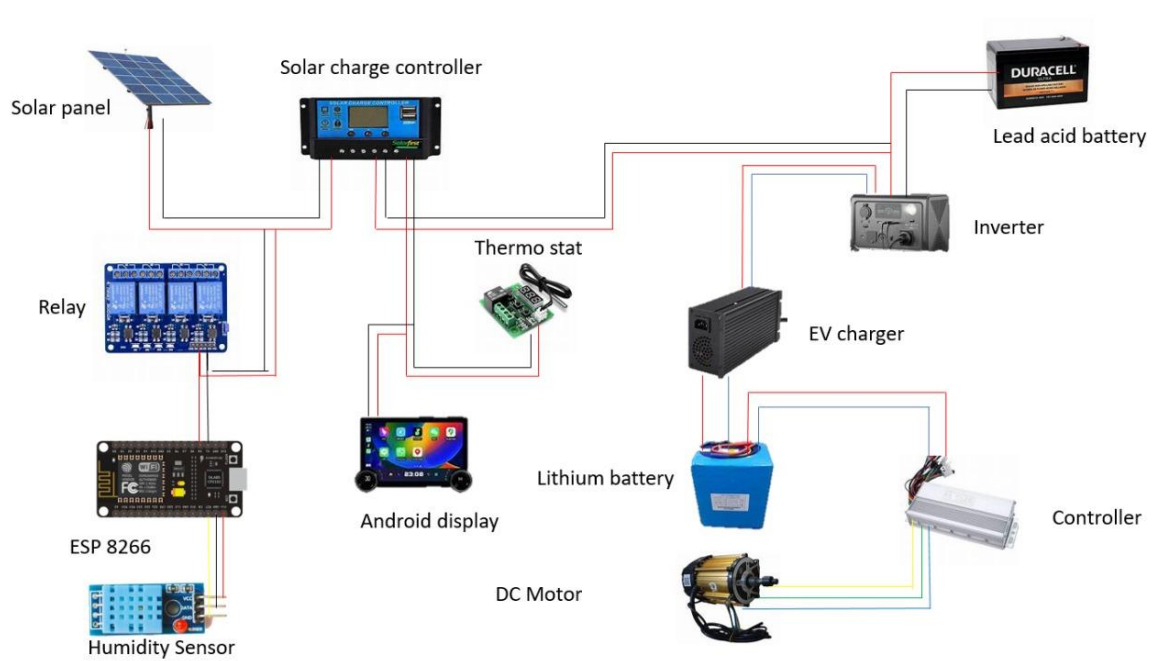
```
void setup()
```

```
{  
  Serial.begin(115200);  
  Blynk.begin(auth, ssid, pass);  
  pinMode(D5,OUTPUT);  
  pinMode(D6,OUTPUT);  
  pinMode(D7,OUTPUT);  
  pinMode(D8,OUTPUT);  
}  
  
void loop()  
{  
  Blynk.run();  
}
```

blynk



Circuite connection





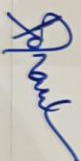
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of ENGINEERING WINNER
has participated in PROJECT DISPLAY
at GIST, National Level Technical Symposium held at Dhaanish Ahmed
College of Engineering on 10/04/2025




Convener


Director



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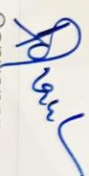


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